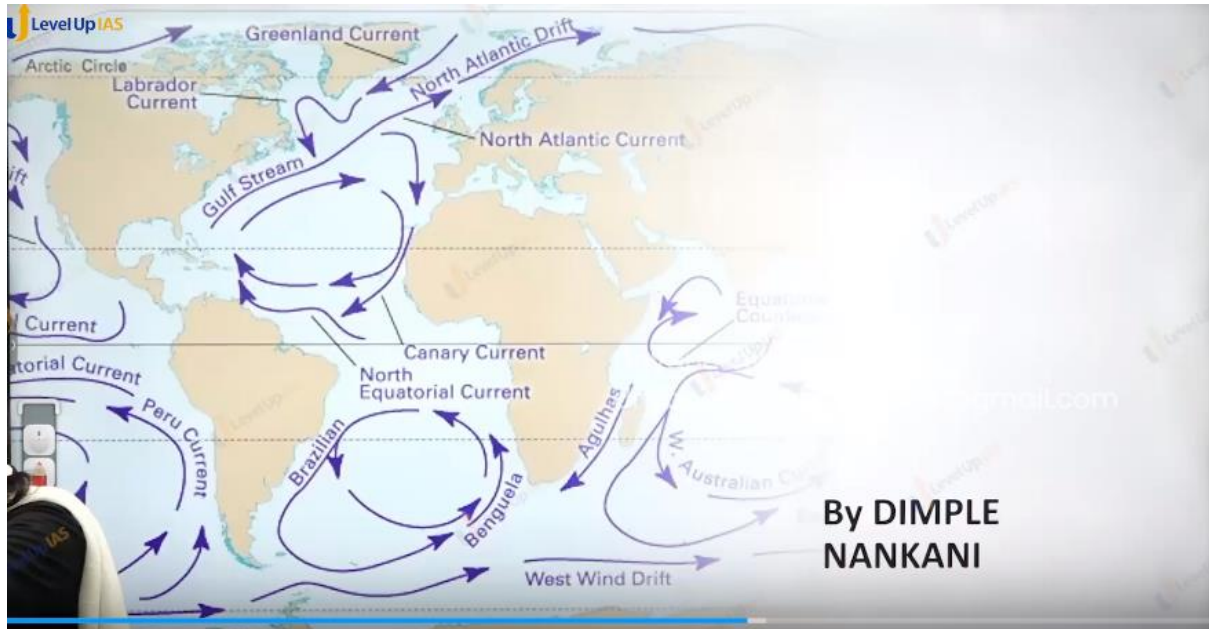
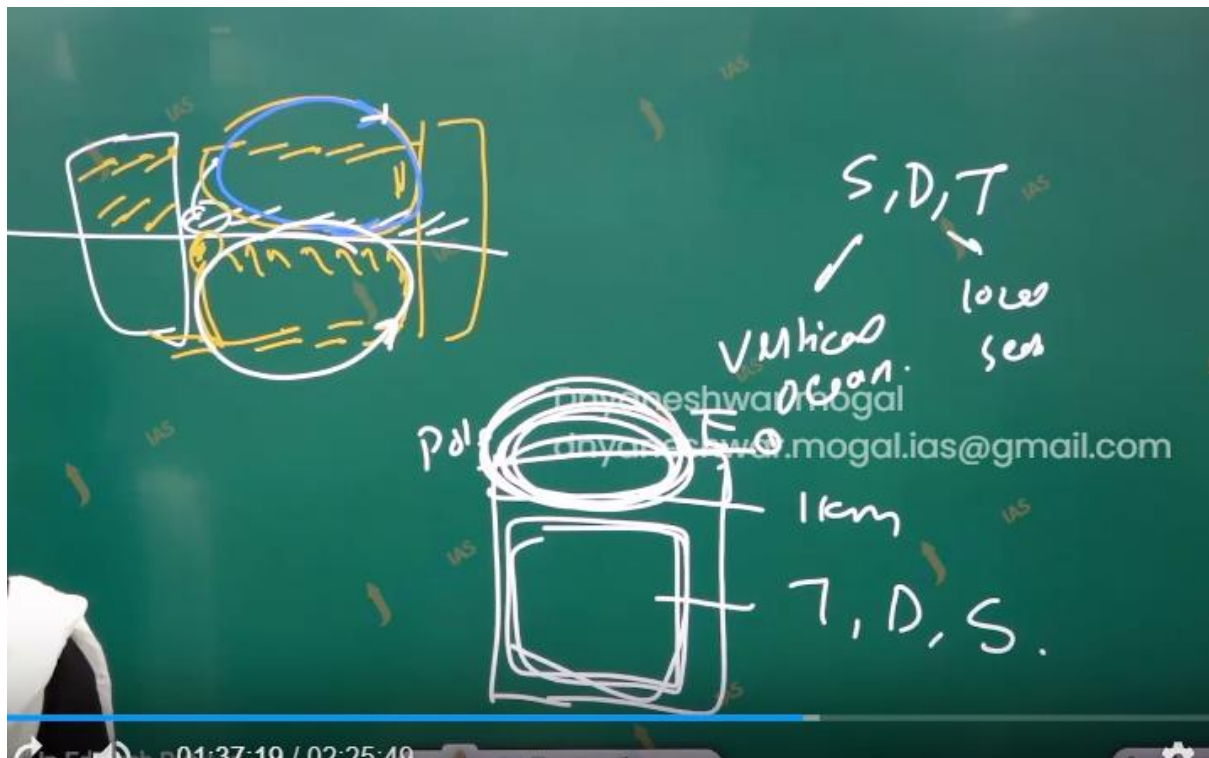


(25) Ocean Currents



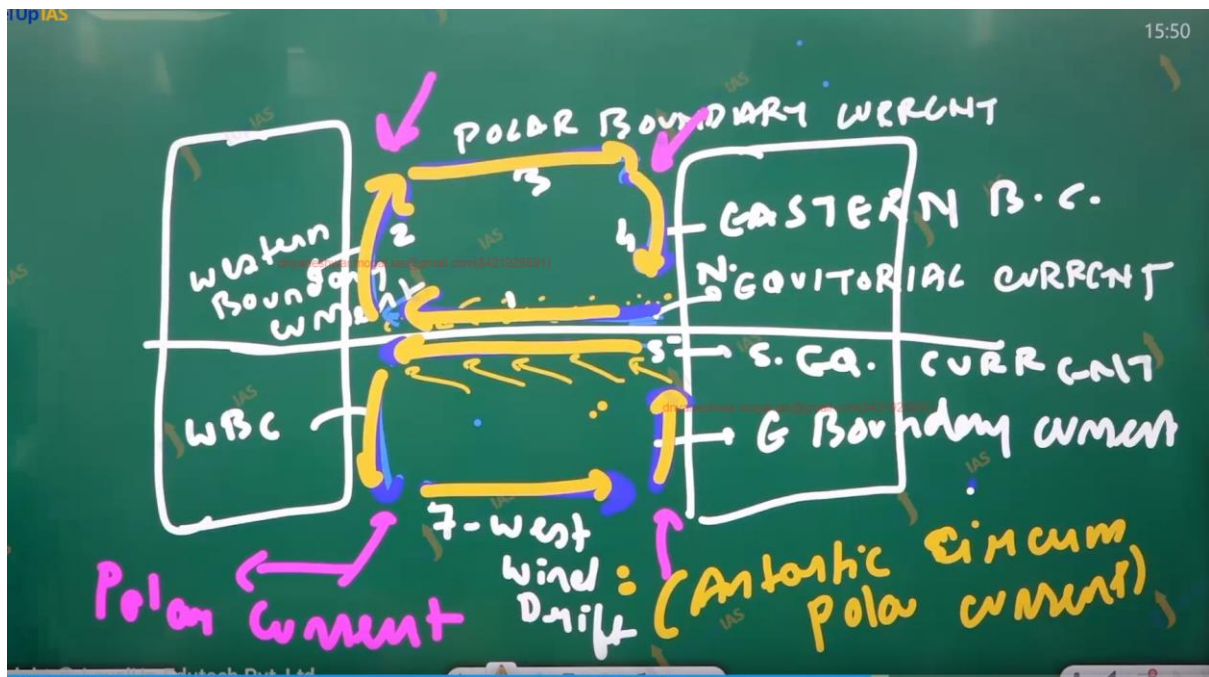
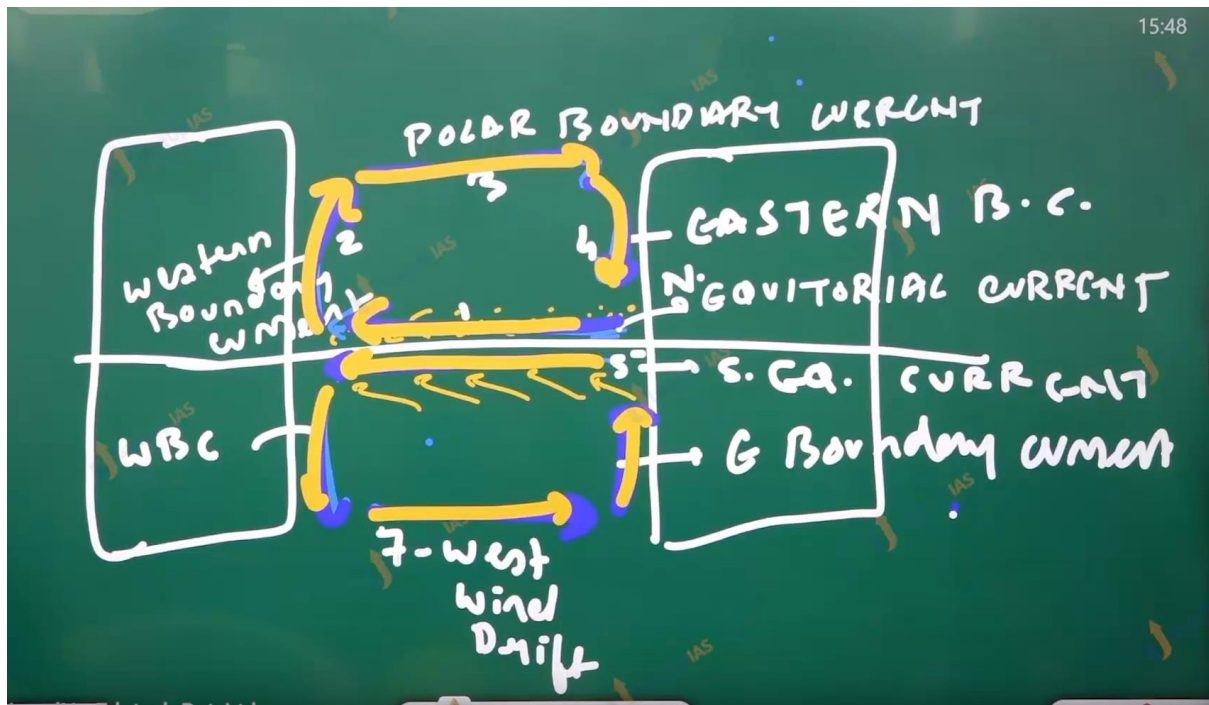
Ocean Currents:

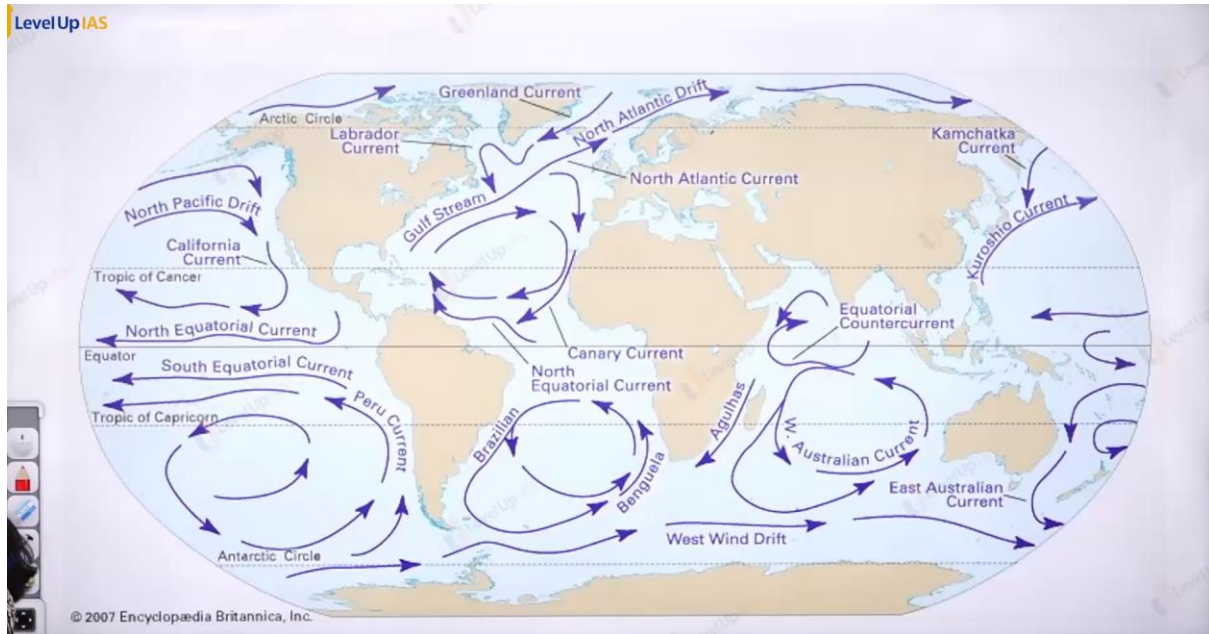
1. Ocean currents refer to the movement of ocean water in a definite direction. They are also called narrow bands of circulate water along the coast.
2. The primary force that generates the ocean currents is wind.
3. Other factors that play an important role are density differences because of salinity and temperature difference.
4. Coastline and ocean bottom topography, ocean bottom reef.
5. Coriolis force



Factor affecting ocean current:

1. Winds and ocean current: trade winds drag the water from east to west. And westerlies drag water west to east.
2. The coastline turns the water to the right in the northern hemisphere. Along with Coriolis force and left in the southern hemisphere.
- 3.





You must learn the how to draw the world map.



Diagram of ocean current and their name.



LevelUp IAS

Ocean Current and Winds

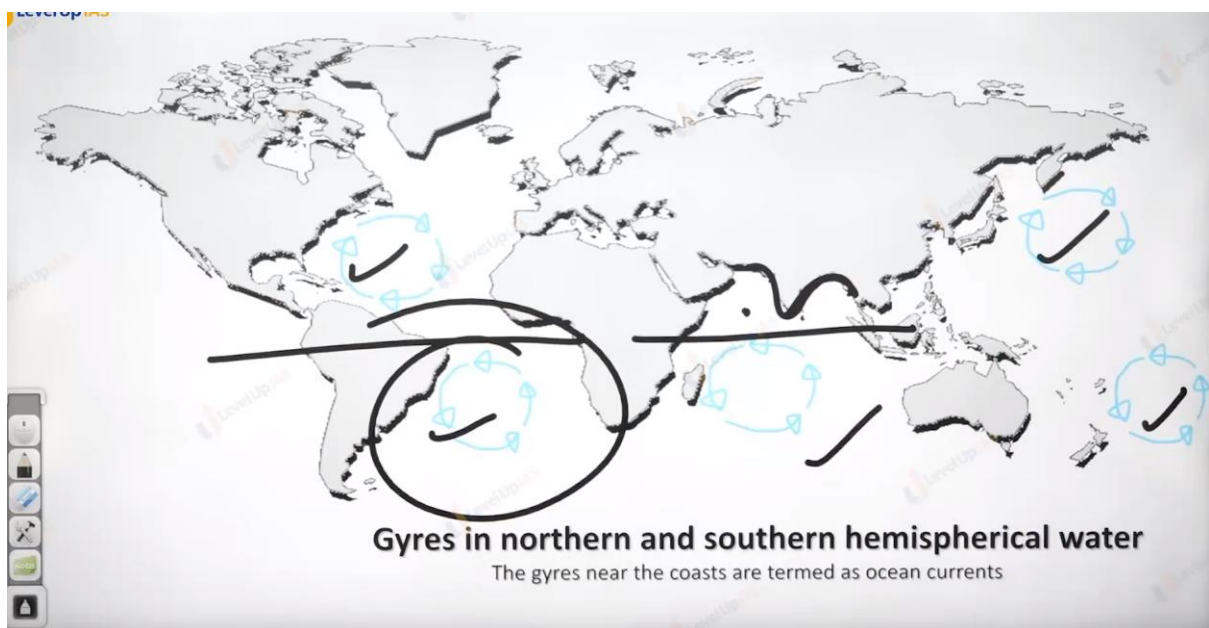
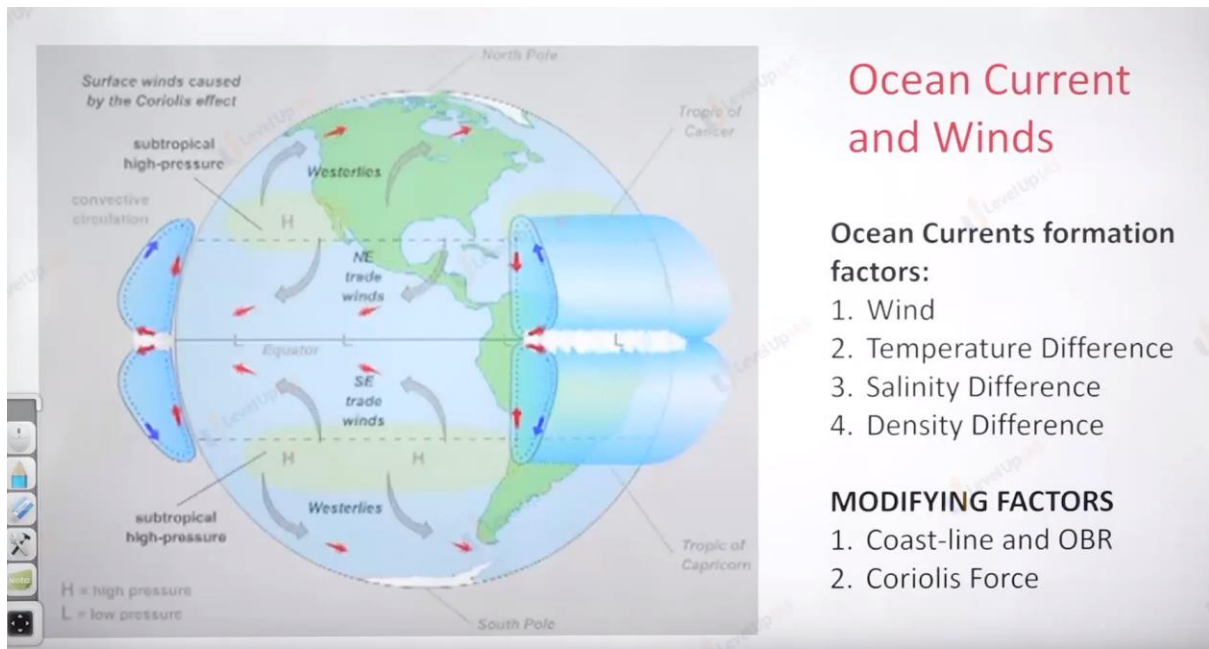
Ocean Currents formation factors:

1. Wind
2. Temperature Difference
3. Salinity Difference
4. Density Difference

MODIFYING FACTORS

1. Coast-line and OBR
2. Coriolis Force

Diagram illustrating the idealized low-latitude convection pattern, disregarding the influence of the Coriolis effect. The diagram shows the Earth with latitude lines (North Pole, Tropic of Cancer, Equator, Tropic of Capricorn, South Pole) and pressure zones (H for high pressure, L for low pressure). Arrows indicate the direction of convective circulation, showing how wind-driven surface currents and density-driven bottom currents interact to form a closed loop.





LevelUp IAS Warm currents of N Atlantic ocean

North Equatorial Current

- Flows east to west between 10° N to 20° N.
- It forms the southern side of the clockwise sub-tropical gyre.

Antilles Current

- It flows past the island chain that separates the Atlantic Ocean and the Caribbean Sea

Florida Current

- Flows around Florida Peninsula and joins the Gulf Stream at Cape Hatteras.
- Discovered in 1513 by Spanish explorer Juan Ponce de León.

Gulf stream

- Western intensified current-driven mainly by wind stress.
- It splits into North Atlantic Drift (crossing Northern Europe & southern stream) and Canary Current

Norwegian Current

- This wedge-shaped current is one of the two dominant Arctic inflows of water.
- It is a branch of North Atlantic Drift and sometimes also considered as an extension of the Gulf Stream.



LevelUp IAS Warm currents of N Pacific ocean

Kuroshio Current

- Also known as Japan Current or Black Current. The term "Kuroshio" in Japanese means "Black Stream".
- It is the Pacific analogue of Gulf Stream.
- The average surface temperature of this current is warmer than the surrounding ocean. This helps in regulating the temperature of Japan, which is relatively warmer.

North Pacific Current

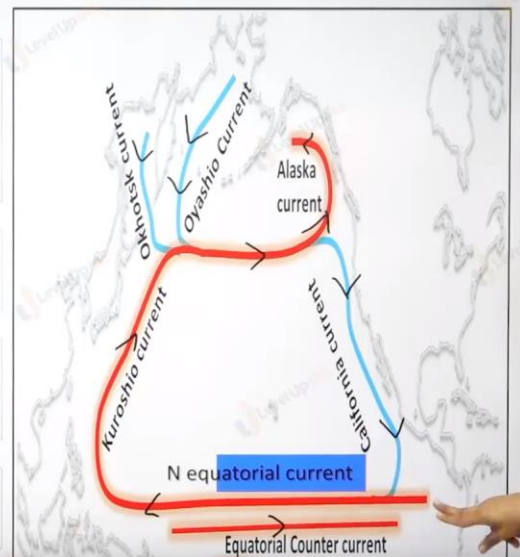
- It is formed by the collision of Kuroshio & Oyashio.
- It circulates counter-clockwise along the Western North Pacific Ocean.

Alaskan Current

- It results from a northward diversion of a part of the North Pacific Ocean.
- It forms two large eddies- Haida Eddies and Sitka Eddy

North Equatorial Current

- Flows east to west between 10° N to 20° N.
- It forms the southern side of the clockwise sub-tropical gyre.



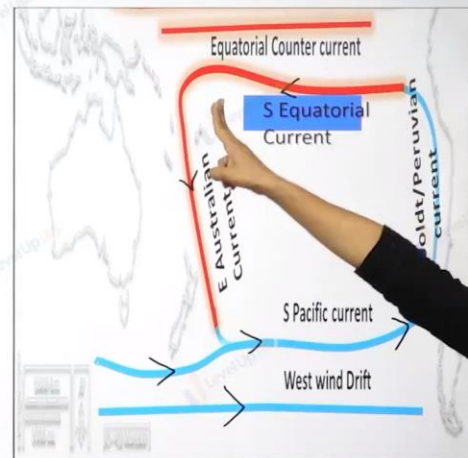
LevelUp IAS Cold currents of S Pacific ocean

Humboldt /Peruvian Current

- Flows from the southernmost tip of Chile to northern Peru, along the west coast of South America.
- This is responsible for scanty rainfall in Peru.
- This low-salinity current has a large marine eco-system serves as one of the major nutrient systems of the world.

Antarctic Circumpolar Current

- It is the largest ocean current in the world.
- It is also known as **West Wind Drift**.
- It flows from east to west around Antarctica in a clockwise direction.



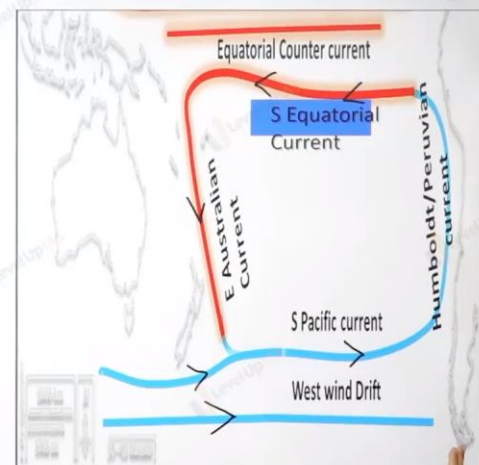
LevelUp IAS Warm currents of S Pacific ocean

East Australian Current

- Acts to transport tropical marine fauna to habitats in sub-tropical regions along the southeast Australian coast.

S Equatorial Current

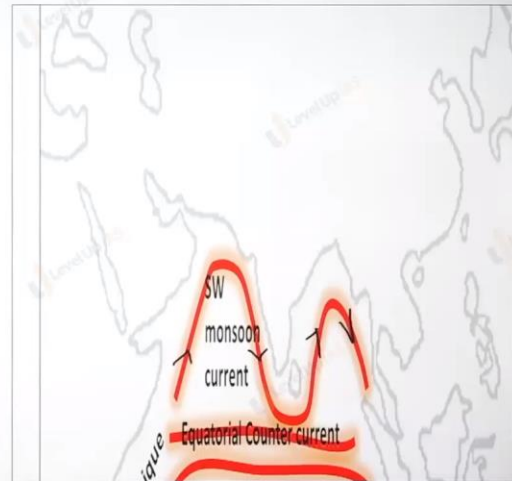
- Directly driven by trade winds blowing from east to west.
- Southern hemisphere counter-part of North Equatorial countercurrent.



Warm currents of N Indian ocean : Summer

Southwest Monsoon Current

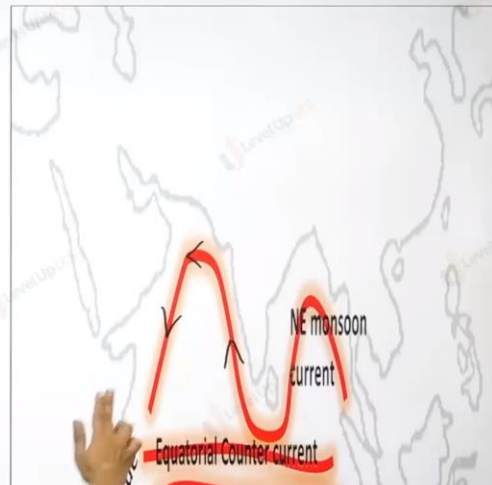
- It dominates the Indian Ocean during the southwest monsoon season (June–October).
- It is a broad eastward flowing ocean current that extends into the Arabian Sea and Bay of Bengal.



Warm currents of N Indian ocean : Winter

Northeast Monsoon Current

- North-east monsoon winds blow from land to the ocean during winter season in the northern hemisphere and thus westward blowing north-east monsoon currents are produced between Andaman and Somali.
- it current flows to the south of 5°N latitude.
- In the Indian Ocean, the summer currents are more dominant than those of the winter.



Currents of S Indian ocean

Mozambique current

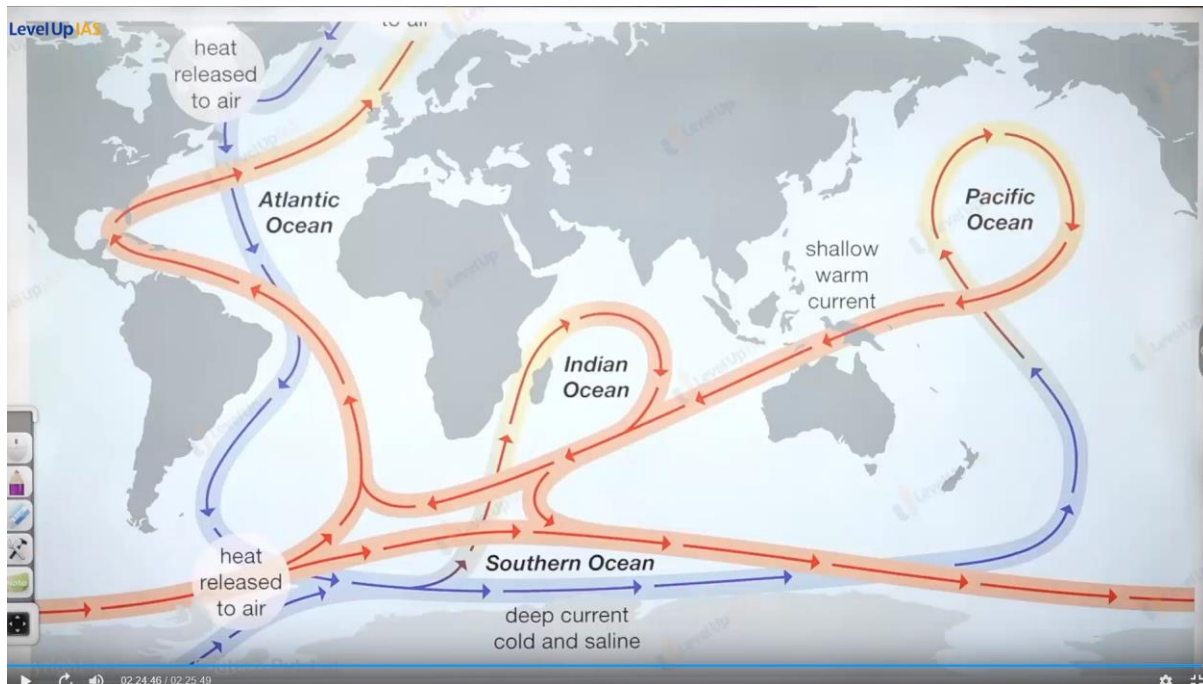
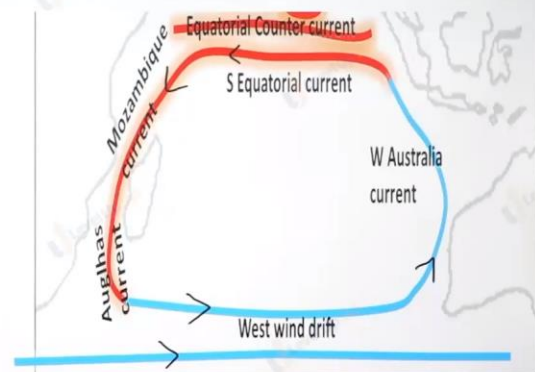
- One branch of the southern Indian Ocean
- equatorial current which moves southward through Mozambique Channel.
- it current joins the Agulhas current near 30°S latitude
- it moves upto the southern tip of Africa and is ultimately diverted eastward.

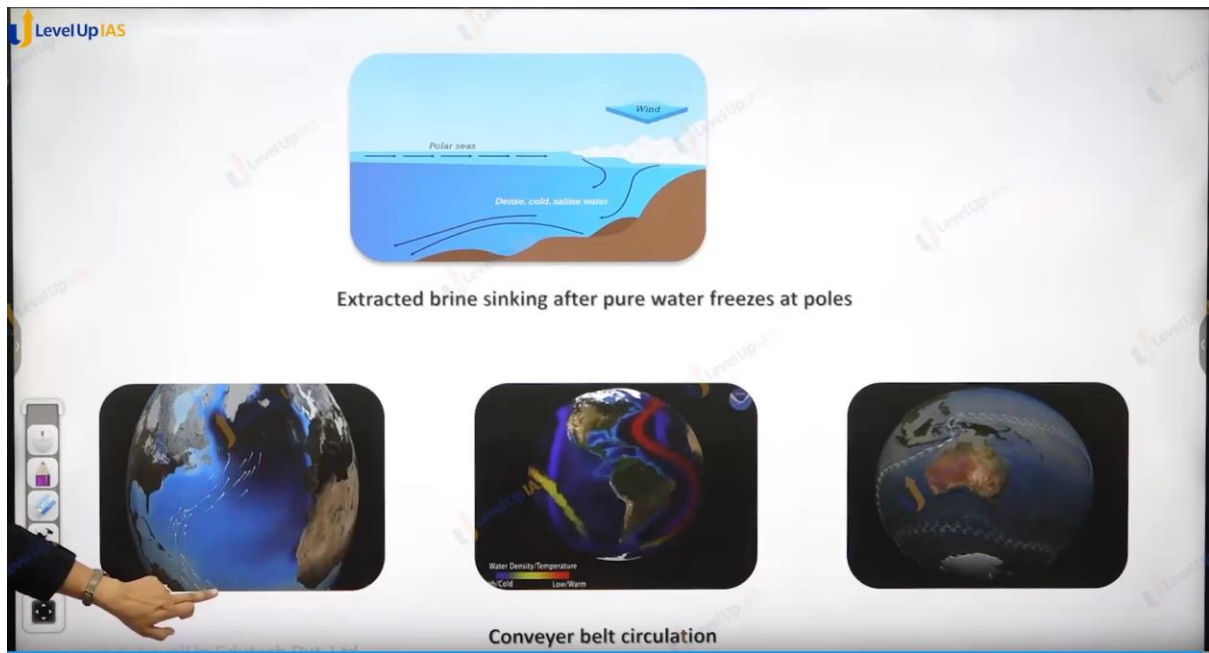
Agulhas current

- It is formed by the confluence of the warm Mozambique and East Madagascar Currents,
- It Flows south along the east coast of South Africa.

W Australia current

- It is the branch of West wind drift bifurcating due to west Australian peninsular tip.
- It is strong in summer and weak in winter.
- It is Cold current.





LevelUp IAS

2012

Consider the following factors:

1. Rotation of the Earth
2. Air pressure and wind
3. Density of ocean water
4. Revolution of the Earth

Which of the above factors influence the ocean currents?

(a) 1 and 2 only (b) 1, 2 and 3
(c) 1 and 4 (d) 2, 3 and 4

Mains Question

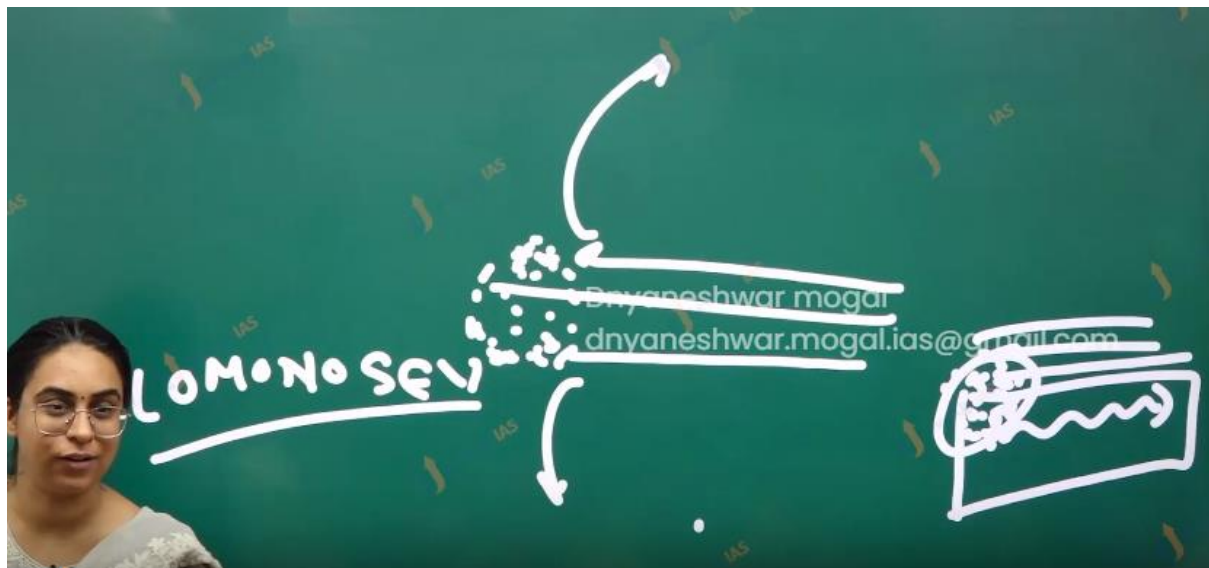
Q) What are the forces that influence ocean currents? Describe their role in fishing industry of the world. **2022 UPSC PYQ**

Q) How do ocean currents and water masses differ in their impacts on marine life and the coastal environment? **2019 USPC PYQ**

Q) Explain the factors responsible for the origin of ocean currents? How do they influence regional climates, fishing, and navigation? **2015 UPSC PYQ**

Q) Most of the unusual climatic happenings are explained as an outcome of the El-Nino effect. Do you agree? **2014 UPSC PYQ**

(26) El Nino



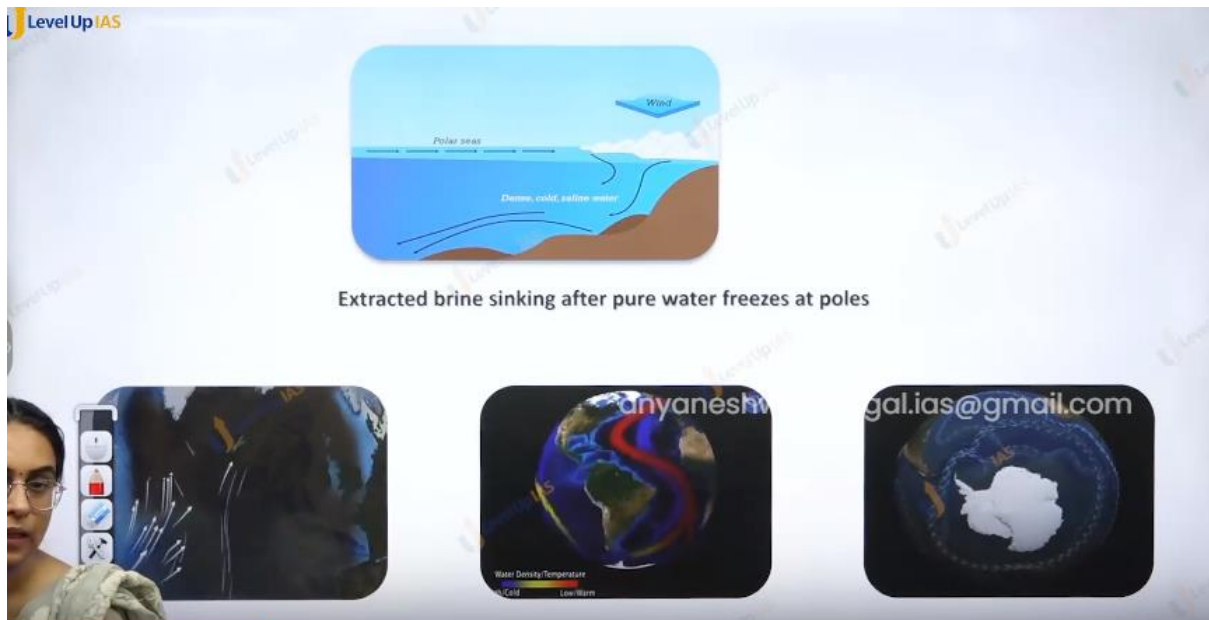
Formation of counter equatorial current:

1. At the equator due to piling up the water because of north and south equatorial current) a counter equatorial current close in the opposite direction that is west to east just below the surface, therefore counter equatorial current is sub-surface current.
2. It is a warm equatorial current. Ex. Lomonosov is counter equatorial current in the Atlantic ocean, Cromwell is CEC in the pacific ocean.,
3. TREEV is CEC of Indian ocean.

North Indian ocean current:

1. The relationship between ocean current and winds best seen in the Indian ocean
2. Due to reversal of winds, there is reversal in the current in the north Indian ocean. Northeast monsoon winds from the land drive waters from the way of Bengal and Arabian sea. In an anticlockwise manner.

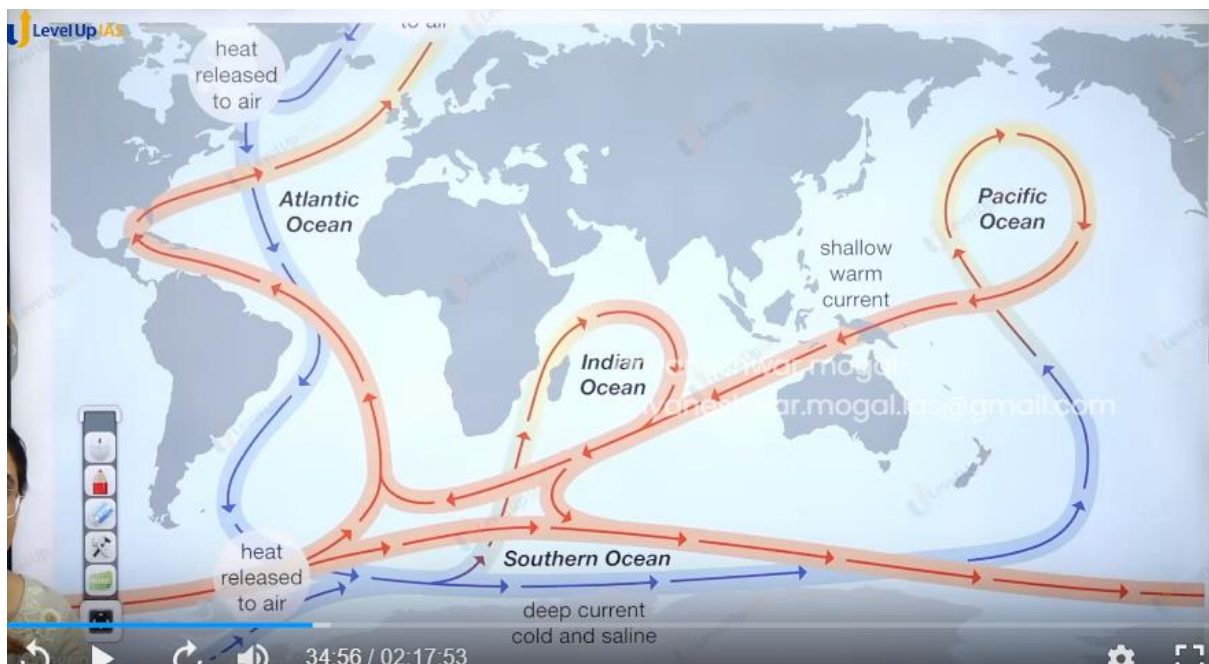
3. Called as north east monsoon drift where as south west monsoon winds produce clockwise flow called as south west monsoon drift.
-



Vertical ocean current: (thermohaline current)

1. Most of the currents are perceptible on the surface, these are surface ocean current.
2. Surface ocean currents is primarily driven by the winds. The effect of winds can be most effective up to 1000m.
3. Deeper water beyond 1000m are not driven by winds but it is driven by temperature difference and salinity water and the density of water.
4. These deep currents are called as stenohaline current.
5. This is large scale oceanic circulation, and they cover 90% oceanic water circulation.
6. It begins near the polar region where ice form the leaving the water saltier and denser leading the sinking of water.

7. This water moves slowly in the single belt / continuous belt across the different ocean and acts as medium to transfer the heat across globe.
 8. Thermohaline circulations are nothing but the movement of water masses.
 9. The water mass huge body of water with nearly same temperature salinity, comparable density.
-



Significance of ocean current:

1. Cold ocean currents along the western coast in the sub-tropical region. Suppresses the precipitation increasing the aridity of desert.
2. Ex Namib desert and Benguela current, Atacama Desert and Peru current. Great Australian desert cold west Australian current. Mohave desert, cold California current. Sahara Desert and cannabiss current.
3. Cold ocean current
4. Warm oceanic current along the east coast in the tropical and subtropical latitude can induce precipitation. The warm ocean along

the temperate regions can create mild winters. And keeps the coast ice free.

5. Ex Norwegian current keeping the port of Norway ice free.
 6. Polar current can reduce temperature of region in the mid latitude.
 7. Ex. Labrador current, east Greenland current.
 8. Meeting upon the warm current and cold current can create optimum temperature current for the growth of plants leading to good fishing ground ex. Great bank of Newfoundland near the Labrador current and Gulf Stream in the north Atlantic ocean and Kuroshio (location north of Japan).
 9. Meeting up warm and cold ocean current can lead to fog formation which can obstruct navigation.
 10. Ocean currents also facilitate navigation.
 11. Ocean currents help in global distribution of temperature and salinity.
 12. Cold ocean currents due to upwelling can increase the nutrients facilitate / provide the good fishing ground.
 13. Ex. Humboldt along the Peruvian region.
-

Importance of water mass:

-
1. It can help in the generation and maintenance of thermohaline circulation.
 2. It can help in redistribution of oxygen salinity and temperature across the ocean
 3. It can help in generation of vertical circulation and facilitate redistribution of heat from the surface to depth and vice versa.
-

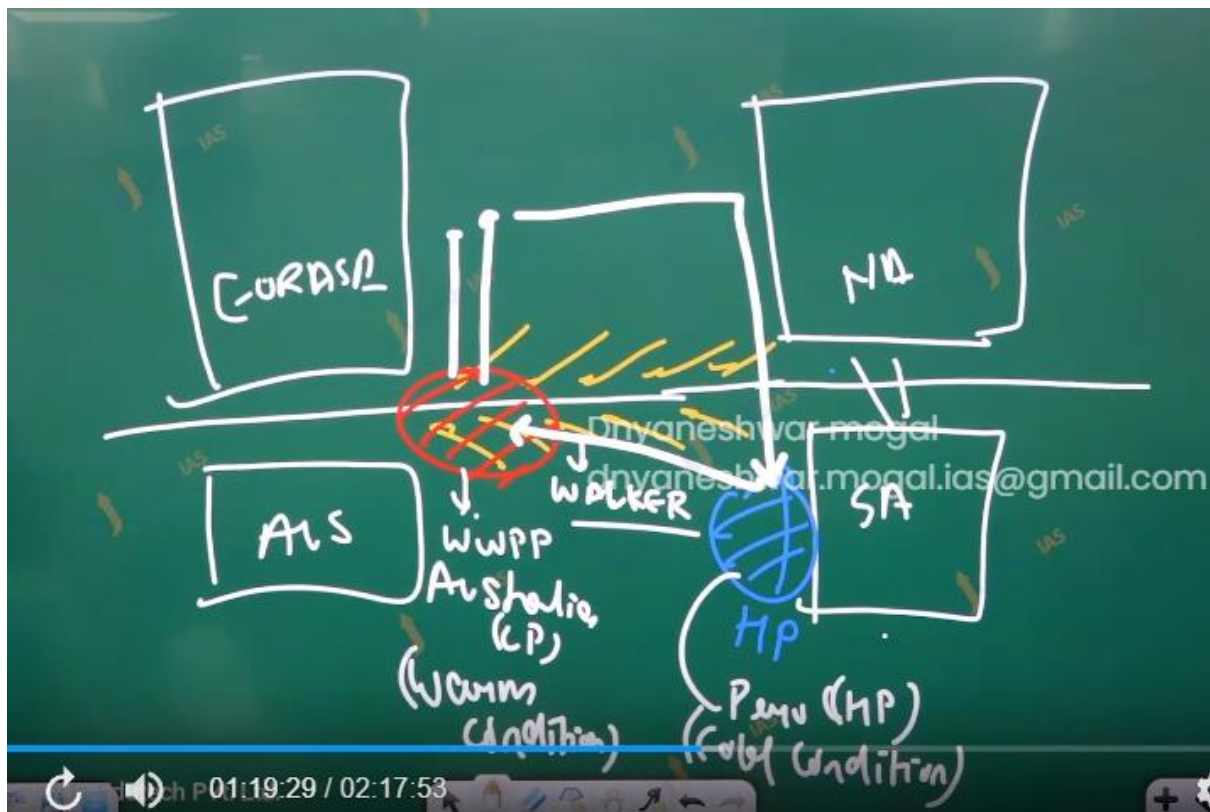
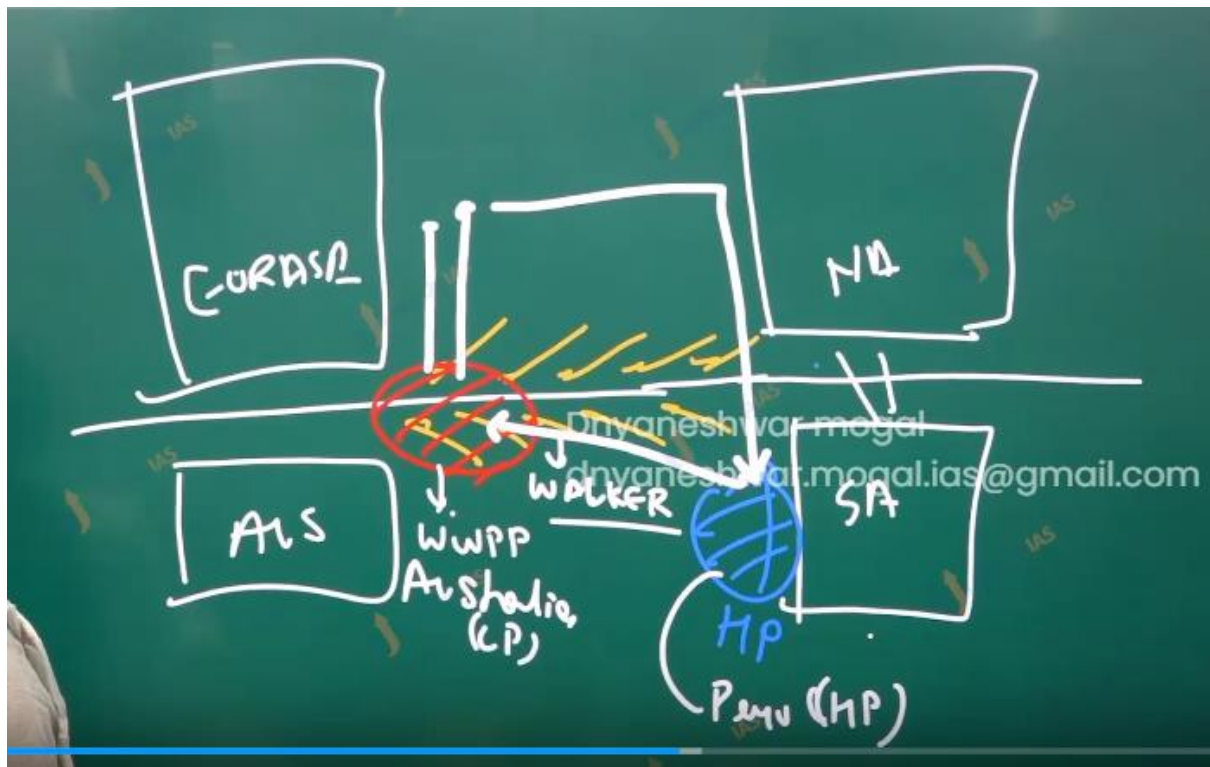
El Nino El Nina //// ENSO

(El Nino southern oscillation).

-
1. Enso is a phenomenon of reversing the ocean currents and reversing the water circulation that can also reverse the atmospheric circulation.
 2. This is associated with alternate flood and drought conditions.
 3. Thos Enso is best developed in the pacific ocean (southern ocean) but it has global impact therefore Enso is called as global ocean atmosphere telecommunication.
-

IN normal years

-
1. Trade winds are very strong.
 2. Equatorial current of pacific are strong
 3. It creates warm wester pacific pool in the western pacific therefore the difference between temperature and pressure conditions exists in the western and eastern pacific in the southern hemisphere there is raising air along the western pacific and sinking air along the eastern pacific. (along Peru coast) this is called as walker circulation
 4. The consequences are warm wet conditions in Indonesia, north of western Australia and dry conditions along Peru coast. That facilitates the good upwelling and good fishing ground.
-



During the El Nino year.

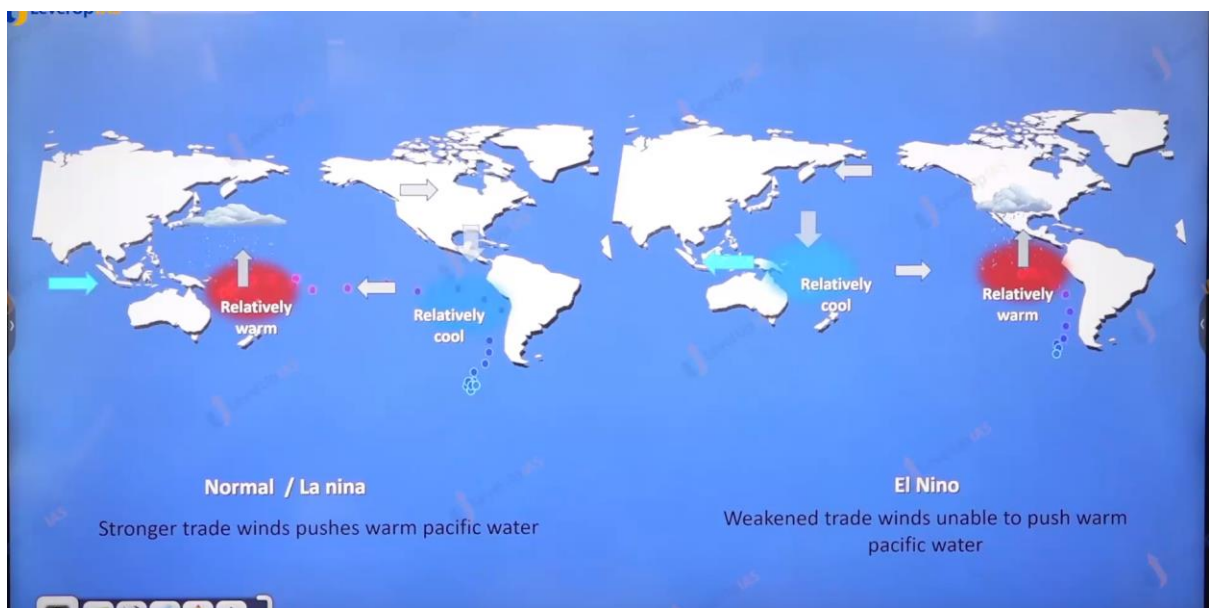
-
1. For the reasons does not know trade winds of the pacific can slow s downs.
 2. This weekend the equatorial current of pacific. Therefore, warm west pacific pools spread backwards and moves eastwards, therefore low pressure in Australia can weaken and may also be converted to high pressure, the high pressure at Peru can weaken or coveted to low pressure.
 3. Therefore, walker cells can weaken or can get reversed.
 4. This leads to drought and flood; it leads to drought and forest fire in Indonesia and Australia and it leads to flood in Peru
 5. It impacts Peru economic because of lack of upwelling and lack of fish catch.
-

La Nina years

-
1. In some years, Tradewinds become unusually strong. Leading to very strong equatorial current, very strong warm wettest pacific pools, very strong Peru currents, very strong walker cells
 2. (Strengthening of normal years.)
 3. Consequences are Indonesia and Australia can have very strong rainfall and floods whereas Peru has very strong upwelling therefore there is very good fish catch. The market will glut. Because of over supply in the market prices crash.
-

El Niño	Normal Year	La Niña
1. Trade winds are very weak	1. Trade winds are strong	1. Trade winds are very strong
2. Eq. current are weak.	2. Equatorial currents are strong.	2. Eq. current are very strong
3. Warm west Pacific pool - not well developed move eastward.	3. Warm west Pacific pool is well developed	3. WWP is very well developed.

3. Warm west Pacific pool - not well developed move eastward.	3. Warm west Pacific pool is well developed	3. WWP is very well developed.
4. upwelling @ Peru - weak	4. upwelling @ Peru - Strong	4. upwelling @ Peru is v. strong
5. Australia has drought & F.F and Peru - flood	- Australia - good rainfall Peru - dry cond ⁿ .	5. Australia - flood Peru - extreme dry cond ⁿ



2012

Consider the following factors:

1. Rotation of the Earth
2. Air pressure and wind
3. Density of ocean water
4. Revolution of the Earth

Which of the above factors influence the ocean currents?

- (a) 1 and 2 only (b) 1, 2 and 3
(c) 1 and 4 (d) 2, 3 and 4

Answer B

Ocean is not stable water bodies. There are motions

-
1. Horizontal circulation motion due to trade winds.
 2. Vertical circulation due to density, salinity, temperature.
 3. wave
 4. Tide (amount of water depends on the revolution.) gravitational pull of sun and moon.
 5. cyclone
-

2015

What explains the eastward flow of the equatorial counter-current?

- (a) The Earth's rotation on its axis
- (b) Convergence of the two equatorial currents
- (c) Difference in salinity of water
- (d) Occurrence of the belt of calm near the equator

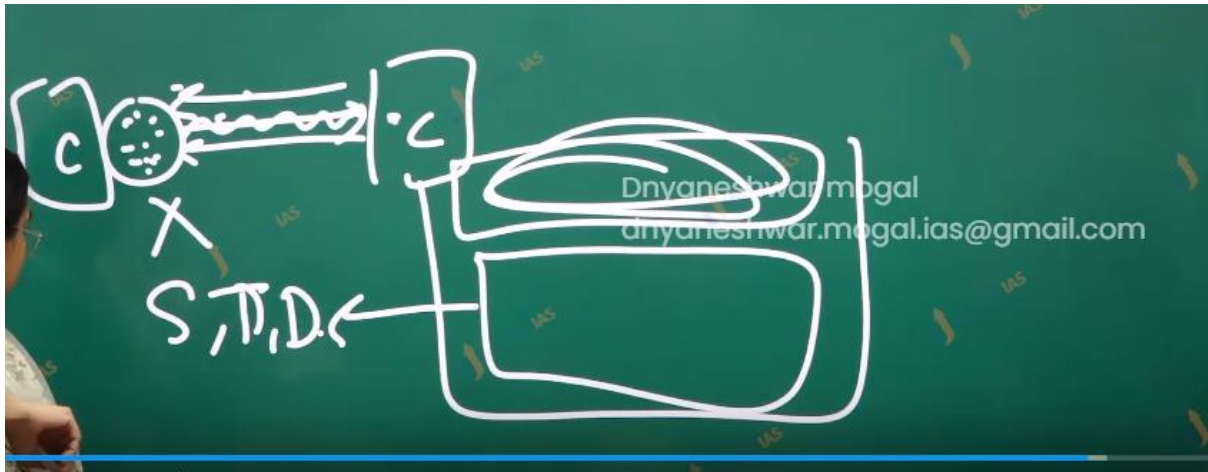
Ans b

Salinity at both points would be the same just water is displaced. Vertical currents are dependent on the difference in salinity. It will create the ocean current when there is enclosed sea.

Belts of calm near the equator can be created due to warm and cold ocean current. Which only affects the weather.

A is wrong because the direction of counter equatorial current is opposite to what it supposed to be. Earth rotates from west to east.

Current should flow from east to west. But we can see their direction is west to east.



Mains Question

Q) What are the forces that influence ocean currents? Describe their role in fishing industry of the world. **2022 UPSC PYQ**

Q) How do ocean currents and water masses differ in their impacts on marine life and the coastal environment? **2019 USPC PYQ**

Q) Explain the factors responsible for the origin of ocean currents? How do they influence regional climates, fishing, and navigation? **2015 UPSC PYQ**

Q) Most of the unusual climatic happenings are explained as an outcome of the El-Nino effect. Do you agree? **2014 UPSC PYQ**

2011

La Nina is suspected to have caused recent floods in Australia. How is La Nina different from El Nino?

1. La Nina is characterised by unusually cold ocean temperature in equatorial Indian Ocean whereas El Nino is characterised by unusually warm ocean temperature in the equatorial Pacific Ocean.
2. El Nino has adverse effect on south-west monsoon of India, but La Nina has no effect on monsoon climate.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans is d

Cold ocean current +++ no rain

Warm ocean current rain.

In A opposite are given that why this is wrong.

In the option second both El Nino and La Nina as effect on monsoon clement.

2015

Tides occur in the oceans and seas due to which among the following?

1. Gravitational force of the Sun
2. Gravitational force of the Moon
3. Centrifugal force of the Earth

Select the correct answer using the code given below.

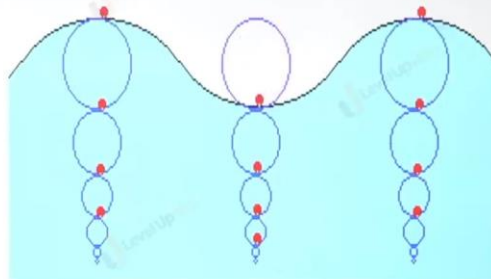
- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3





Figure 14.2 : Relation between gravitational forces and tides

Waves



Waves:

1. Waves are disturbances or ripples formed over seas surface as energy moves from one place to another.
2. Do not involve movement of water particles it is only the energy which moves.
3. Formed when producing force remains smaller than the resisting force due to which the energy is unable to carry the water particle and is only successful in creating disturbances

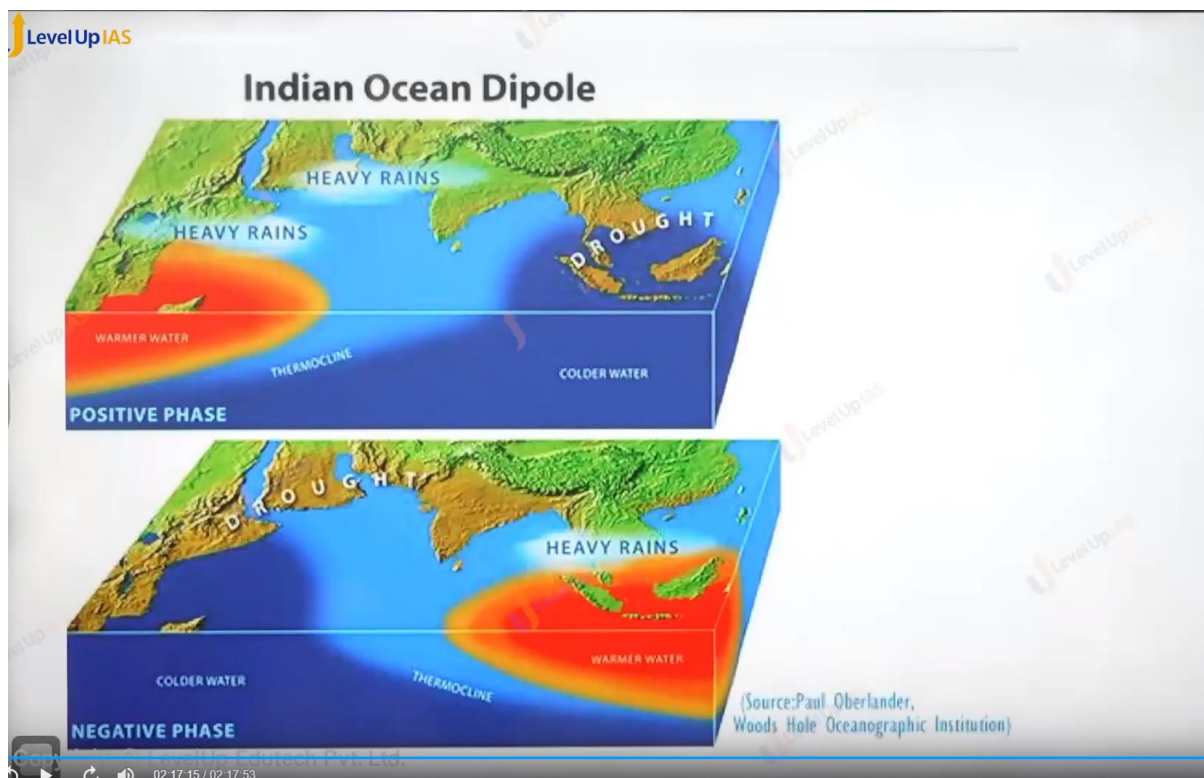
Ans d .

With reference to “Indian Ocean Dipole (IOD)” sometimes mentioned in the news while forecasting Indian monsoon, which of the following statements is/are correct?

1. IOD phenomenon is characterised by a difference in sea surface temperature between tropical Western Indian Ocean and tropical Eastern Pacific Ocean.
2. An IOD phenomenon can influence an El Nino’s impact on the monsoon.

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor



LevelUp IAS
2018

Consider the following statements

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and the Philippines.
3. Coral reefs host far more animal phyla than those hosted by tropical rainforests.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 3 only
(c) 1 and 3 only
(d) 1, 2 and 3



02:17:16 / 02:17:53

LevelUp IAS
2018

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